Check for updates

Received: 18 May 2018 | Revised: 26 February 2020 | Accepted: 19 March 2020

DOI: 10.1002/joom.1089

RESEARCH ARTICLE

WILEY

Common institutional investors and supplier performance in supply chains

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Handling Editor: Fabrizio Salvador

[Correction added on 28 July 2020 after first online publication: Sriram Narayanan and Subodha Kumar were incorrectly included as handling editors.]

Abstract

We investigate how common institutional investors (CIIs) in supply chains affect supplier performance. Social network theory suggests that buyer–supplier relationships are influenced by networks of ties in which they are embedded. While prior research has concentrated on networks of trade interactions, we instead examine the influences of networks through common investors. Utilizing a large sample of buyer–supplier relationships, we find that the presence of CIIs improves suppliers' operating and market performance, especially for suppliers with greater dependence on buyers. In addition, supplier performance increases with the common ownership stakes held by CIIs, but decreases with the asymmetric ownership stakes of CIIs across supply chain partners. We find that better financial collaboration between the partners appears to be a mechanism through which CIIs enhance supplier performance. Lastly, we find that the role of CIIs in strengthening supplier performance is distinct from that of direct equity links. Collectively, our findings highlight a novel role of indirect cross-ownership in fostering supply chain collaboration and coordinating vertical relationships and, in turn, improving supplier performance.

KEYWORDS

asymmetric ownership stake, buyer–supplier relationships, common institutional investors (CIIs), common ownership stake, supplier performance

1 | INTRODUCTION

Buyer–supplier relationships are influenced by the networks in which they are embedded. In addition to influences related to the characteristics of a specific buyer–supplier relationship, the perceptions and actions of both partners are influenced by structural and relational attributes stemming from the direct and indirect ties they have with other partners, associations, and institutions (Cao & Lumineau, 2015; Dong, Ju, & Fang, 2016; Wang, Zhang, Wang, & Sheng, 2016). Researchers have created a sizable and growing literature using social network theory, social exchange theory, and social capital theory to explain how social ties fostered by network embeddedness create the potential for governance mechanisms, as well as for informational and reputational advantages (Dong, Liu, Yu, & Zheng, 2015; Kim, 2014; Moran, 2005). To date, however, research has neglected an important network in which buyer–supplier relationships are embedded—the network of ties created by common institutional investors (CIIs) who simultaneously own shares in buyers and suppliers.[1]

The present study examines the influences of CIIs shared by buyer–supplier dyads. Because of the low redeployability of relationship-specific investments in a supply chain, both suppliers and buyers have incentives to

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exploit the other partner. In such a setting, direct or indirect ownership linkages establish effective governance mechanisms to safeguard investments and create value for both partners. Whereas a buyer or supplier who partially owns its partner is inclined to maximize its own value, a common investor for a buyer–supplier pair seeks to maximize the joint value of the portfolio (Ahmadjian & Oxley, 2005; Lanier Jr, Wempe, & Zacharia, 2010; Williamson, 1985).[2] As a result, supplier and buyer policies in the presence of CIIs could significantly differ from those absent CIIs, especially those policies that create externalities for the other partner. For instance, partners with CIIs are likely to view relationship-specific investments differently from those without. In a more general setting, a growing body of literature has established that common owners seek to maximize their portfolio values by inducing their portfolio firms to internalize externalities (e.g., Hansen & Lott, 1996; Rubin, 2006).

Despite the important role of CIIs, there is little understanding on how CIIs influence value creation in buyer–supplier interactions. Social network theory suggests that embedded ties created by intense co-investments by CIIs mitigate frictions inherent in buyer–supplier relationships. Common investors likely exploit their access to buyer and supplier information to influence the decisions of both parties. Such influence might be applied through informational, reputational, monitoring, and/or controlling elements (Kim, 2014) designed to limit opportunism and maximize the combined value of both parties (Heide, Wathne, & Rokkan, 2007; Kim, 2014). Prior research shows that institutional investors can influence firms through direct communications with top managers (McCahery, Sautner, & Starks, 2016). CIIs can also directly make shareholder proposals aimed at improving supply chain performance and integration (He, Huang, & Zhao, 2019). In addition, CIIs can nominate directors to the buyer–supplier pairs to mitigate supply chain frictions (Rubin, 2006). For example, CIIs can link the executive compensation of one partner with the performance of the other and thus increase the incentive of executives to collaborate to improve joint performance of the partners (Liang, 2016). As such, common investors might push both parties to invest in interorganizational systems to decrease information asymmetries and to improve coordination (Sanders, 2008; Wang & Wei, 2007). Beyond direct influences, the presence of CIIs may encourage the development of relational norms that breed stronger information sharing, cooperation, and trust between buyers and suppliers. A recognition of their common institutional owners can prompt buyer–supplier pairs to engage in joint value creation, while curbing opportunism (Granovetter, 1985; Gulati & Sytch, 2007).

In this study, we draw upon social network theory to develop and test predictions regarding the effects of CIIs on supplier performance. Specifically, we investigate the effects of the presence of CIIs and the levels of common ownership stake and asymmetric ownership stake of CIIs on supplier performance. In addition, we examine the role of a supplier's asymmetric dependence on the buyer as a moderator of the beneficial effects of CII, and a mechanism through which CIIs affect supplier performance. Finally, we compare the distinct effects of direct and indirect equity links in supply chains.

Based on a large sample of buyer–supplier relationships and a research design that decomposes the effect of CIIs into between- and within-firm components, we find that the presence of CIIs is positively associated with the operating and market performance of supplier firms. This effect is more pronounced when the suppliers are more dependent on the principal buyers. Further, we find that supplier firm performance increases with common ownership stakes but decreases with asymmetric ownership stakes held by CIIs across the buyer–supplier dyad. These results are mostly driven by the between-firm components of the common (asymmetric) ownership stakes of CIIs and are consistent with our conjecture that CIIs foster collaboration, enhance coordination, and curb opportunism in buyer–supplier interactions, thereby improving supplier performance. Our findings are corroborated by a battery of robustness tests addressing endogeneity concerns, using alternative performance measures, and comparing long-term versus short-term ownership. We also identify trade credits as a mechanism through which CIIs influence supplier performance. Finally, we find that the effect of indirect equity links through CIIs is distinct from that of direct equity links in the supply chain.

This study makes several contributions to the literature. First, while a number of studies evaluate the performance effects of direct equity linkages in a buyer–supplier dyad (e.g., Allen & Phillips, 2002; Dyer, 1997; Fee, Edward, Hadlock, & Thomas, 2006; Gerlach, 1992), we focus on the indirect ownership created through CIIs. In the corporate world, indirect common ownership through CIIs is far more prevalent than direct ownership in supply chains. Fee et al. (2006) document that only 3.31% of buyer–supplier relationships (in a sample of 10,211 buyer–supplier-years) have direct equity linkages. In our preliminary sample (46,353 buyer–supplier-years),[3] however, 42% of buyer–supplier relationships have indirect equity linkages through CIIs. Our findings indicate that the positive influences of CIIs on the supplier performance are distinct from, and more substantial than, those of direct equity links in buyer–supplier dyads. Furthermore, we find that the beneficial effects of CIIs increase with suppliers' asymmetric dependence on the major buyers. This finding extends our understanding of the interaction between network effects and power dependence.

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Second, this study documents an economically significant *net* benefit of CIIs in creating value in the supply chain; 1 *SD* increase in the between-firm component of the presence of CIIs corresponds to a 2-percentage-point rise in suppliers' gross profit margin. Further, we find that supplier firm performance increases with common ownership stakes but decreases with asymmetric ownership stakes held by CIIs across a buyer–supplier dyad. Therefore, by directing attention to the performance implication of equity link through CIIs, this study offers a novel extension of prior social network studies on supply chains, which largely focus on material flows (Kim, Choi, Yan, & Dooley, 2011) and knowledge and information flows (Bellamy, Ghosh, & Hora, 2014). Third, the exploratory portion of our study identifies trade receivables as a mechanism through which CIIs influence supplier performance, suggesting fruitful opportunities for future research. Lastly, this study provides an original comparison of the distinct effects of direct and indirect equity links in the supply chain, extending prior work on the role of direct ownership (e.g., Dyer, 1997; Fee et al., 2006). Altogether, our findings are important to suppliers, buyers, and investors who seek to understand the role of institutional cross-ownership in creating value in vertical relationships.

2 | LITERATURE REVIEW AND HYPOTHESIS DEVELOPMENT

Buyers and suppliers are linked by incomplete contracts and relationship-specific investments that vary in level and specificity (Williamson, 1996). Because of differences in dependence and redeployability of investments, an exchange partner, either a supplier or a buyer, often has incentives to exploit quasi-rent (Handley & Benton, 2012). Effective governance mechanisms can alleviate such opportunism and improve collaboration between the parties (Cao & Lumineau, 2015; Williamson, 1975).

Structural embeddedness of buyers and suppliers within social networks fosters informational, reputational, monitoring, and controlling influences, which act as governance mechanisms (Kim, 2014). For example, social network theory suggests that firms embedded in a network of interdependent relationships develop relational norms that breed stronger information sharing, cooperation, and trust, thereby facilitating joint value creation while curbing opportunism (Granovetter, 1985; Gulati & Sytch, 2007). Prior research has explored the influences of embeddedness in buyer–supplier networks (Dong et al., 2015; Kim, 2014; Kim, 2017) and institutional (mostly national) networks (Cao & Lumineau, 2015; Dong et al., 2016; Handley & Angst, 2015; Sheng, Zhou, Li, & Guo, 2018; Wang et al., 2016). The present study seeks to advance social network theory by applying it to the context of ownership networks.

Prior studies find that institutional investors have information advantages and play monitoring roles in corporate activities (Kochhar & David, 1996). A varied stream of research documents evidence regarding the effects of CIIs on the outcome of mergers and acquisitions (Matvos & Ostrovsky, 2008), financial reporting (Jung, 2013), and industry competition (Azar, Schmalz, & Tecu, 2018; He & Huang, 2017). Matvos and Ostrovsky (2008) show that the distorted incentives of common owners lead to adverse consequences of mergers and acquisitions. Jung (2013) finds that a firm tends to follow a first-mover in information disclosure if the firm and the first-mover have common owners. Recent studies find that CIIs in the same industry affect industry competition. Azar et al. (2018) document that common investors in the airline industry strengthen industry market power, leading to higher ticket prices. In contrast, He and Huang (2017) find that CIIs in the same industry strengthen the coordination of product market decisions, leading to higher market share growth for the commonly held firms.

Closer to our interests, Hansen and Lott (1996) demonstrate that common owners seek to maximize their portfolio values by inducing their portfolio firms to internalize externalities. Similarly, Rubin (2006) claims that common investors push boards of directors to make decisions based on the total benefit of firms within the investment portfolios, rather than the benefits of individual firms. These studies suggest that common owners maximize joint portfolio value as opposed to individual firm value. However, this stream of research addresses collections of firms typically selected by owners to minimize risk through diversification. Externalities and interdependencies among firms who populate such portfolios are broad and indirect, and perpetuated through common industry membership, competition, or complementarities. Our study of common ownership in buyer–supplier dyads provides a much more granular analysis of CIIs' influences on firms who are directly tied together through specific commercial trade agreements and relationship-specific investments.

In a concurrent study, Freeman (2019) finds that common institutional ownership along supply chains increases the relationship duration and transaction size. Though both studies investigate the economic consequences of CIIs in buyer–supplier dyads, our study is distinct from Freeman (2019) in three important ways. First, we focus on the aggregate economic effects of CIIs in buyer–supplier dyads, which are captured by the operating and market performance of suppliers. As such, our findings corroborate and extend those from Freeman (2019). Second, we demonstrate that the common ownership stakes

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and asymmetric ownership stakes of CIIs exhibit conflicting effects on supplier performance, and thus extend Freeman (2019) who examines only one dimension of the effects. Third, we present novel evidence on the relative effects of direct and indirect cross-ownership on supply chain relationships over and beyond Freeman (2019) who examines only the effect of the latter.

We propose that embeddedness in networks of common ownership creates structural ties that influence buyer and supplier behaviors in three primary ways: fostering collaboration, enhancing coordination, and curbing opportunism. First, CIIs foster co-commitment and norms of solidarity, that is, the willingness to strive for joint benefits (Rokkan, Heide, & Wathne, 2003). Institutional investors' interactions and communications with buyers and suppliers encourage them to view their relationship as a single maximizing unit as opposed to exchange between separate maximizers (Macneil, 1981). Activities (such as relationship-specific investments) of one partner often have externalities to the other. Common owners can internalize such externalities and raise the expectation of repeated exchange, thereby inducing partners to collaborate to maximize their joint benefits (Dyer, 1996, 1997; Jap, 1999). For instance, CIIs can link the executive compensation of one partner with the performance of the other, which strengthens the incentive of collaboration to enhance joint performance of the partners (Liang, 2016). More broadly, a shared vision of the future through common ownership encourages collaboration (Dyer, 1997; Handley & Benton, 2012; Rokkan et al., 2003).

Second, common investors can access and share information from both the supplier and buyer (Cohen & Frazzini, 2008), enabling them to incorporate financial and operating conditions of a buyer–supplier pair into decisions that could maximize their joint value (Heide et al., 2007; Kim, 2014). CIIs create structural channels for information processing and coordination (Bensaou & Venkatraman, 1995). For example, investors can hold private discussions with executives or board members of the partners, providing them an opportunity to influence the managerial and operating decisions of the partners (McCahery et al., 2016).

Third, CIIs may provide structures for curbing opportunism arising in the buyer–supplier interactions. For example, CIIs can nominate directors to both buyers and suppliers to mitigate supply chain frictions (Rubin, 2006). As noted above, such structural embeddedness leads to relational embeddedness and ties that increase the costs of opportunism and/defections by either party (Carey, Lawson, & Krause, 2011; Kim, 2014; Parkhe, 1993; Schelling, 1978). In addition, relational norms foster cooperation and impose sanctions on partners who violate norms and shared values (Liu, Luo, & Liu, 2009; Polidoro, Ahuja, & Mitchell, 2011; Zaheer & Bell, 2005). Firms utilize relational norms as safeguards against opportunism and to protect relationship-specific investments (Heide & John, 1988; Lee & Johnson, 2010; Lui & Ngo, 2004).

Studies on buyer–supplier relationships in the United States generally find that suppliers tend to be more bound by the relationship than buyers (Chen, Zhang, & Zhang, 2016; Hertzel, Li, Officer, & Rodgers, 2008), they are more dependent, and they have lower bargaining power (Cen, Maydew, Zhang, & Zuo, 2017; Radhakrishnan, Wang, & Zhang, 2014). Therefore, we expect the influences of CIIs in fostering collaboration, facilitating coordination, and curbing opportunism to be particularly beneficial for suppliers, as they often occupy the weaker position in the relationship. Accordingly, we focus on supplier performance to investigate the influences of CIIs in buyer–supplier relationships. Our first hypothesis builds on the expectation that CIIs strengthen vertical relationships and mitigate potential conflicts in buyer–supplier dyads, leading to favorable performance of suppliers:

Hypothesis 1 *When CIIs are present in buyer–supplier dyads, suppliers are more profitable than otherwise.*

Our second hypothesis relies upon the power dependence theory, which posits that the higher-power party uses its power to appropriate value from the lower-power party (Emerson, 1962). The literature contends that a supplier's dependence on a given buyer grows in proportion to the fraction of the supplier's total sales that are consumed by the buyer (Elking, Paraskevas, Grimm, & Corsi, 2017; Kim, 2017; Kim & Zhu, 2018; Lanier Jr et al., 2010). Similarly, a buyer's dependence grows in proportion to the fraction of its total purchases provided by the supplier.

A number of researchers argue that asymmetric dependence in a buyer–supplier relationship, namely the differential levels of dependence of one partner on the other, creates power imbalance which encourages opportunism, primarily because the dependent (weaker) party lacks abilities to influence the independent (stronger) party (Casciaro & Mikolaj, 2005; Elking et al., 2017; Kim & Zhu, 2018; Park & Ungson, 1997). Thus, asymmetric dependence benefits the higher-power party, but harms the lower-power party in the short term. A number of prior studies provide related empirical evidence. Crook and Combs (2007) find that when task interdependence among supply chain members is not reciprocal, strong members utilize their bargaining power to appropriate partnership gains. Using the relative dependence of a distributor on the manufacturer as a proxy for potential hold-up in supply chains, Schloetzer (2012) finds that a decrease

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in asymmetric dependence improves process integration and information sharing between partners in the supply chain, leading to favorable financial performance for the partners and contract renewal. Park and Ungson (1997) document how dependence asymmetries lead to instability and dissolution in joint ventures. Kim and Zhu (2018) show that suppliers' lack of monitoring ability and financial distress caused by heavy dependence on major buyers limit their abilities to invest in R&D.

The foregoing discussions suggest that asymmetric dependence exerts adverse effects on the lower-power party in a buyer–supplier relationship, as it is likely to face opportunistic behaviors and threats from the powerful counterpart. In such a case, CIIs who seek to maximize their portfolio value likely have stronger incentives to promote joint improvement efforts, create channels for coordination, and curb opportunism, thereby defusing frictions between the parties and improving collaboration (Bensaou & Venkatraman, 1995; Benton & Maloni, 2005; Zhao, Huo, Flynn, & Yeung, 2008). Taken together, we expect that the beneficial influence of CIIs is more pronounced for suppliers with greater asymmetric dependence on their major buyers. Therefore, our second hypothesis is stated as follows:

Hypothesis 2 *The effect of the presence of CIIs on the supplier's profitability is more pronounced when the supplier is more asymmetrically dependent on the buyer.*

Next, we consider the impacts of two distinct features of ownership that CIIs hold in the partners. The first feature is the level of common ownership, defined as the minimum ownership stake (i.e., percentage of shareholding) a CII holds in either the supplier or buyer in a supplier–buyer dyad. When CIIs hold a higher level of common ownership stake across the two partners, they have a greater incentive to maximize their portfolio value by strengthening collaboration, improving coordination, and curbing opportunism by the powerful partner (Kochhar & David, 1996; Wilson, 1975). In contrast, institutional investors with a lower level of common ownership stake have limited abilities to influence the supply chain partners, as they might be superseded by owners with greater shareholdings. Accordingly, we expect the hypothesized effect in H1 to increase with the level of common ownership stake of CIIs. Our third hypothesis is, therefore:

Hypothesis 3 *The positive effect of the presence of CIIs on the supplier's profitability increases with the level of common ownership of CIIs in a supplier–buyer dyad.*

Moreover, we expect that asymmetry in ownership stakes held by CIIs in a supplier–buyer dyad matters. We define asymmetric ownership (ownership asymmetry) as the absolute difference in the ownership stakes held by a CII across the supplier and buyer. If CIIs hold an equal ownership stake in a buyer and its supplier, they aim to maximize the joint value of the pair and have little incentive to transfer profit from one to the other (Hansen & Lott, 1996). However, if their stakes in the partners are asymmetric, common investors have an incentive to transfer profit from the partner with a lower stake to the one with a higher stake to maximize the portfolio value. As such, CIIs with higher asymmetry in ownership stakes between partners have a lower incentive to strengthen collaboration, foster coordination, and curb opportunism in the supply chain. Thus, the benefits of CIIs for suppliers are muted by the asymmetric ownership stakes. This leads to our fourth hypothesis as follows:

Hypothesis 4 *The positive effect of the presence of CIIs on the supplier's profitability decreases with the level of asymmetric ownership of CIIs in a supplier–buyer dyad.*

3 | DATA AND VARIABLES

3.1 | Buyer–supplier data

We select a sample period over 1982–2011 in order to obtain all required data from various data sources. Following the literature, we identify buyer–supplier relationships using COMPUSTAT Segment Customer files.[4] First, to get the unique buyer identifiers, we use an algorithm to match buyer names to the name file of COMPUSTAT, and select the one with the smallest difference. Then, we conduct a visual inspection and use industry information to verify whether the selected one is indeed the buyer for a given supplier. Finally, we match the buyer identifiers to identifiers in COMPUSTAT, CRSP, and Thomson Reuters, and obtain fundamental buyer information and institutional shareholding information. To exclude cases where the supplier and buyer are in a business group, we use data from WRDS corporate library to check whether the supplier (buyer) is the dominant shareholder of the buyer (supplier), or they have the same dominant shareholder. Since WRDS corporate library data begin from 2001 and have limited coverage, we also use the WRDS blockholders' database and the SEC EDGAR system[5] to identify blockholders of a supplier and buyer and delete those buyer–supplier fiscal years where the shareholding percentage by one over the other is greater than 25%.

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We exclude observations where either the supplier or buyer is a financial (SIC codes 6000–6999) or utility firm (SIC codes 4900–4999), observations where supplier sales or buyer cost of goods sold is negative and where the proportion of sales to the buyer or the proportion of cost of goods sold from the supplier exceeds 1. We also drop observations with missing values for variables employed in the regressions. To distinguish the between- and within-firm components of our explanatory variables,[6] we exclude suppliers with only 1 year of data in the sample period. Our final sample consists of 11,052 supplier firm-years for 14,708 pair-years over the sample period.

3.2 | Institutional investor and other data

Institutional investor data are extracted from the Thomson Reuters Institutional (13F) Holdings database from 1982 to 2011.[7] As Klein and Zur (2009) find that hedge funds, individuals, private equity funds, venture capital firms, and asset management groups are monitors, we focus on a broad group of institutional investors, including investment companies and their managers, and independent investment advisors. Specifically, we consider Type 3 (investment companies and their managers) and Type 4 (independent investment advisors) institutional investors in the Thomson Reuters Institutional (13F) Holdings database. We search through these data to identify institutional investors that hold shares in both the buyer and supplier. A supplier–buyer relationship is deemed to have a CII in year t if an institutional investor simultaneously holds shares of the buyer and supplier for more than two quarters of year t.

Financial data come from COMPUSTAT or CRSP/COMPUSTAT Merged database. Market data are from CRSP/COMPUSTAT Merged database. Data for directors are from RiskMetrics (currently Institutional Shareholder Services [ISS]), and data for analysts are from IBES. All nominal variables are winsorized at the 1% and 99% levels to minimize the effect of outliers.

3.3 | Measuring common ownership, asymmetric ownership, and dependence in buyer–supplier dyads

Cohen and Frazzini (2008) and Jung (2013) measure common ownership by the number of CIIs over the total number of institutional investors. In this study, we measure the average common ownership stake of CIIs ($CII_CO_{i,t}$) for a buyer–supplier relationship i in year t as follows:

$$CII_CO_{i,t} = \sum_{j=1}^{J} CII_CO_{i,t,j} \times Supplier\ ownership_{i,t,j} \quad (1)$$

where j is the number of CIIs for a relationship i. $CII_CO_{i,t,j}$ is measured by a CII j's percentage of shareholding in a buyer or supplier of the relationship i, whichever is lower, in year t (see Figure 1). The lower ownership is considered the lower bound of the CIIs' incentives to improve collaboration and coordination and curb opportunism in the supply chain. $Supplier\ ownership_{i,t,j}$ is the percentage of shares held by a CII j in a supplier of the relationship i in year t. We weight $CII_CO_{i,t,j}$ by supplier ownership as it determines the influence that CIIs can exert on the supplier's operating decisions.[8]

The average asymmetric ownership stake of CIIs (CII_ASY) for the buyer–supplier relationship i in year t is calculated as follows:

$$CII_ASY_{i,t} = \sum_{j=1}^{J} CII_ASY_{i,t,j} \times Supplier\ ownership_{i,t,j} \quad (2)$$

$CII_ASY_{i,t,j}$ is calculated as the absolute difference between the supplier ownership stake and the buyer ownership stake held by a CII j in the relationship i in year t. For buyer–supplier relationships without CIIs, $CII_CO_{i,t}$ and $CII_ASY_{i,t}$ are equal to 0. Figure 1 further illustrates the calculation of $CII_CO_{i,t}$ and $CII_ASY_{i,t}$.

We measure asymmetric dependence ($ADEP_{i,t}$) of a supplier on a buyer as the distance between sales dependence (i.e., supplier dependence on the buyer) based on the percentage of supplier sales to the buyer and cost dependence (i.e., buyer dependence on the supplier) based on the percentage of buyer cost of goods sold from the supplier (Elking et al., 2017; Kim, 2017; Kim & Zhu, 2018; Lanier Jr et al., 2010). Factors such as product complexity, requirements for relationship-specific investments, and resource scarcity (e.g., technological know-how) can also cause asymmetric dependence in buyer–supplier relationships. We consider these factors using an alternative measure of dependence in the later analysis.

Since one supplier may have more than one buyer, we treat all buyers for a given supplier as a portfolio and weight $CII_CO_{i,t}$, $CII_ASY_{i,t}$, and $ADEP_{i,t}$ by the percentage of supplier sales to each buyer to construct supplier firm-level counterparts[9]: $CII_CO_{f,t}$, $CII_ASY_{f,t}$ and $ADEP_{f,t}$. For empirical analyses, we use the decile ranks of $CII_CO_{f,t}$ and $CII_ASY_{f,t}$.

4 | RESEARCH DESIGN

To test our hypotheses, we extend the supplier firm performance model of Patatoukas (2012) to include an

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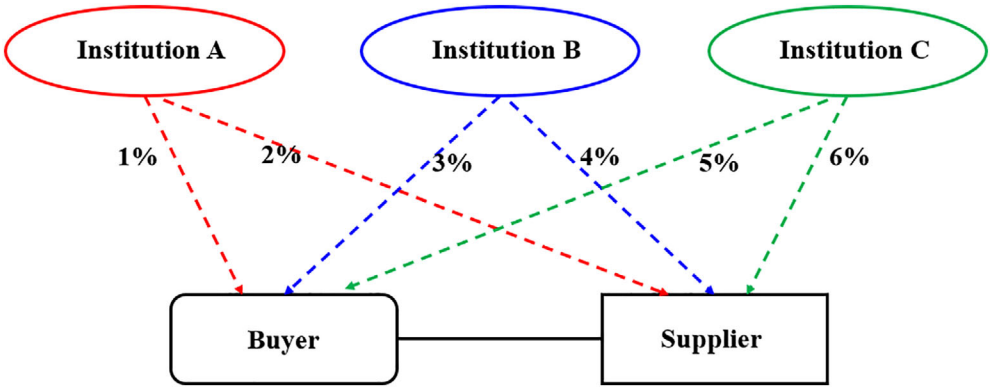

FIGURE 1 Illustration of common institutional ownership along a supply chain. (A)–(C) are institutional investors who simultaneously hold shares of the supplier and buyer. As an example, the figure shows the ownership stakes of each common institutional investor in the supplier and buyer at the end of the last quarter of their shareholdings in year *t*. The common institutional ownership ($CII_CO_{i,t}$) at the relationship level is calculated as min (1%, 2%) × 2% + min (3%, 4%) × 4% + min (5%, 6%) × 6% = 0.46%; the asymmetric institutional ownership ($CII_ASY_{i,t}$) at the relationship level is calculated as |1%–2%| × 2% + |3%–4%| × 4% + |5%–6%| × 6% = 0.12%

indicator of CIIs (*CII_D*), firm-specific institutional ownership (*INST*), and other common governance variables.[10] Institutional investors tend to invest in firms with high performance (Covrig, Lau, & Ng, 2006; Ferreira & Matos, 2008; Lang & Mcnichols, 1997). To mitigate the reverse causality concern, we measure supplier performance in year $t + 1$ (e.g., Balafas & Florackis, 2014; Callen & Fang, 2013; Hall, Mairesse, & Mohnen, 2010; Qian & Steiner, 2017)[11] and control for supplier performance in year *t*. The extended model is specified as follows:

$$\begin{aligned} GPM_{ft+1} = {} & \alpha_0 + \alpha_1 CII_D_{ft} + \alpha_2 CDI_{ft} + \alpha_3 CAN_{ft} \\ & + \alpha_4 CAU_{ft} + \alpha_5 INST_{ft} + \alpha_6 Size_{ft} \\ & + \alpha_7 Sales\ growth_{ft} + \alpha_8 Age_{ft} + \alpha_9 Debt\ ratio_{ft} \\ & + \alpha_{10} ROS_{ft} + \sum\nolimits_j \partial_{1j} FYEAR_j \\ & + \sum\nolimits_j \partial_{2j} IND_j + \omega_{ft} \end{aligned} \tag{3}$$

where *f* indexes supplier firms and *t* indexes years. *GPM* is the gross profit margin of suppliers, calculated as net sales minus cost of goods sold divided by net sales. We use the gross profit margin as the main operating performance measure because it is most relevant to the buyer–supplier relationships and commonly used as a performance measure for buyer–supplier relationships (e.g., Corbett & Karmarkar, 2001; Kulp, Lee, & Ofek, 2004; Radhakrishnan et al., 2014). In the following sections, we also employ alternative operating and market performance measures.

CII_D_{ft} captures the existence of CIIs. We first measure $CII_D_{i,t}$ at the relationship level, in which it takes a value of 1 if a relationship has at least one CII, and 0 otherwise. Then, we weight the relationship-level $CII_D_{i,t}$ by the corresponding percentage of sales to each buyer to obtain the counterpart at the supplier firm level, CII_D_{ft}. Likewise, we construct CDI_{ft}, CAN_{ft}, and CAU_{ft}, which capture the existence of a common director, a common analyst, and a common auditor, respectively. We control for these variables to account for the effects of other communication channels between the supplier and buyer. We also control for institutional ownership ($INST_{ft}$) as prior research finds a positive effect of institutional ownership on firm performance (e.g., Smith, 1996). Controlling for $INST_{ft}$ also allows us to detect the *incremental* effect of CIIs beyond other institutional investors. $Size_{ft}$ is the natural logarithm of total assets. $Sales\ growth_{ft}$ is the growth rate of sales. Age_{ft} is the number of years since the supplier listed or existed in COMPUSTAT, whichever is earlier. $Debt\ ratio_{ft}$ is total debt divided by total assets. Following prior literature (e.g., Waldman, Ramirez, House, & Puranam, 2001), we control for supplier net profit margin in year *t*, measured by net income divided by net sales (ROS_{ft}). *FYEAR* and *IND* are indicators for fiscal years and industries based on two-digit SIC codes. Appendix A provides detailed variable definitions.

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Though our panel data exhibit both within- and between-firm variations along key variables, Equation (3) does not distinguish the between- and within-firm effects of explanatory variables on supplier performance and assumes the two are identical. Certo, Withers, and Semadeni (2017) find that the between-firm relationship differs from the within-firm relationship in the majority of strategic management studies involving panel data, and stress that theorizing and testing the two types of relationships generate additional insights. Therefore, following Certo et al. (2017) and Schunck (2013), we decompose each explanatory variable in Equation (3) into the between- and within-firm components and estimate the following hybrid regression model:

$$\begin{aligned} GPM_{ft+1} = {} & \beta_0 + \beta_1 \overline{CII_D}_f + \beta_2 \left(CII_D_{ft} - \overline{CII_D}_f\right) \\ & + \beta_3 \overline{CDI}_f + \beta_4 \left(CDI_{ft} - \overline{CDI}_f\right) + \beta_5 \overline{CAN}_f \\ & + \beta_6 \left(CAN_{ft} - \overline{CAN}_f\right) + \beta_7 \overline{CAU}_f \\ & + \beta_8 \left(CAU_{ft} - \overline{CAU}_f\right) + \beta_9 \overline{INST}_f \\ & + \beta_{10} \left(INST_{ft} - \overline{INST}_f\right) + \beta_{11} \overline{Size}_f \\ & + \beta_{12} \left(Size_{ft} - \overline{Size}_f\right) + \beta_{13} \overline{Sales\ growth}_{\mathrm{f}} \\ & + \beta_{14} \left(Sales\ growth_{ft} - \overline{Sales\ growth}_f\right) \\ & + \beta_{15} \overline{Age}_f + \beta_{16} \left(Age_{ft} - \overline{Age}_f\right) + \beta_{17} \overline{Debt\ ratio}_f \\ & + \beta_{18} \left(Debt\ ratio_{ft} - \overline{Debt\ ratio}_f\right) + \beta_{19} \overline{ROS}_f \\ & + \beta_{20} \left(ROS_{ft} - \overline{ROS}_f\right) + \sum\nolimits_j \varphi_{1j} FYEAR_j \\ & + \sum\nolimits_j \varphi_{2j} IND_j + \varepsilon_{\mathrm{ft}} \end{aligned} \tag{4}$$

$\bar{X}_f$ is the average value of X_{ft} of firm f over the sample period, denoting the between-firm component of X_{ft}. X_{ft}-$\bar{X}_f$ denotes the within-firm component of X_{ft}. We have no a priori theory to expect the between-firm effect of CIIs to differ from the within-firm effect. Therefore, Hypothesis 1 is supported if β_1 and/or β_2 is significantly positive.

To test Hypothesis 2, we follow Shaver (2019) and use subsample analyses.[12] We split supplier-years by the industry median of $ADEP_{ft}$ into high vs. low dependence groups. Hypothesis 2 predicts β_1 and/or β_2 to be significantly greater in the high dependence group than in the low dependence group.

To test Hypotheses 3 and 4, we add the proxies for the common ownership stake of CIIs (CII_CO_{ft}) and the asymmetric ownership stake that CIIs hold across the partners (CII_ASY_{ft}) to Equation (3) and further decompose them into $\overline{CII_CO}_f$, $CII_CO_{ft} - \overline{CII_CO}_f$, $\overline{CII_ASY}_f$ and $CII_ASY_{ft} - \overline{CII_ASY}_f$. Hypothesis 3 predicts that the coefficient on $\overline{CII_CO}_f$ or $CII_CO_{ft} - \overline{CII_CO}_f$ is significantly positive, while Hypothesis 4 predicts that the coefficient on $\overline{CII_ASY}_f$ or $CII_ASY_{ft} - \overline{CII_ASY}_f$ is significantly negative. To address the concern that within-firm residuals are positively correlated, we use the standard errors clustered by firm to test the significance of coefficients.

5 | EMPIRICAL RESULTS

Current accounting standards (SFAS 14 and SFAS 131) require firms to disclose major buyers (greater than 10% of sales), but not major suppliers. As a result, our sample has comprehensive information about major buyers, but limited information about major suppliers. The sample average (median) sales dependence of suppliers is about 20% (15%) and the average (median) cost dependence of buyers is about 2% (0.2%). Given that the dependence of buyers on their suppliers is quite low, it is unlikely that the sample contains a substantial representation of suppliers who threaten dependent buyers. Hence, the sample composition supports our focus on dependent suppliers in buyer–supplier relationships. In the following sections, we first present the descriptive statistics of sample firms' characteristics and main variables. Then, we provide main results concerning the effects of CIIs on supplier performance. Lastly, we provide a series of robustness tests, including endogeneity checks, and a potential mechanism test.

5.1 | Descriptive statistics

Table 1 presents the firm-level statistics for suppliers and buyers in our final sample. Consistent with prior research, buyers are much larger and older than suppliers and a typical COMPUSTAT firm. Buyers have more capital expenditures (*CAPEX*) and lower growth potential (*TobinQ*) than suppliers on average. However, the mean and median values of R&D intensity (*R&D/TA*) of suppliers are significantly higher than those of buyers. Compared with a median COMPUSTAT firm, suppliers have larger size, higher growth potential, a longer history of listing, and greater R&D expenditures. In total, our final sample (consisting of distinct supplier and buyer firms) accounts for around 12% of COMPUSTAT firms over our sample period, excluding financial and utility industries.

Table 2 reports the descriptive statistics of our regression variables. The average gross profit margin (GPM_{ft+1}) is 24%. Its intraclass correlation coefficient[13] is 0.71 (0.76^2/[$0.76^2 + 0.49^2$]), suggesting that for GPM_{ft+1}, between-firm variance is the primary source of variance. The weighted average value of the presence of CIIs (CII_D_{ft}) is 82%. The greater standard deviation of $\overline{CII_D}_f$ relative to

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TABLE 1 Sample description

		COMPUSTAT (1)	Supplier (2)	Buyer (3)
Total Assets ($ in million)	Mean	1,750	1,987	22,325
	Median	75	178	5,572
	Obs.	215,921	11,052	5,265
CAPEX	Mean	0.08	0.06	0.07
	Median	0.04	0.04	0.05
	Obs.	215,231	11,052	5,265
TobinQ	Mean	14.04	2.22	2.08
	Median	1.45	1.56	1.54
	Obs.	183,724	11,020	5,252
Age	Mean	11.64	15.37	26.74
	Median	8.00	11.00	24.00
	Obs.	232,054	11,052	5,265
R&D/TA	Mean	0.10	0.08	0.04
	Median	0.00	0.03	0.01
	Obs.	215,231	11,052	5,265

Note: The variables in the first column are calculated using the COMPUSTAT data after excluding firm-years from financial and utility industries, and over our sample period. *CAPEX* is capital expenditure divided by total assets. *TobinQ* is calculated as the sum of market value of equity and the book value of total liabilities divided by total assets. *Age* equals fiscal year minus the minimum between IPO year and the first year the firm exists in COMPUSTAT. *R&D/TA* equals R&D expenditure divided by total assets. Missing capital expenditure and R&D expenditure values are set to 0.

that of $CII_D_{ft} - \overline{CII_D_f}$ suggests that the between-firm variance of CII_D_{ft} is greater than its within-firm variance. For the two characteristics of common institutional ownership, the weighted average (median) common ownership stake of CIIs is 0.08% (0.05%), and the weighted average (median) asymmetric ownership stake of CIIs is 0.21% (0.11%).[14] In terms of the dollar amount of investment for CIIs, the average (median) market value of stocks they hold in the partner with lower ownership is around $88 ($13) million, which is about 5% (15%) of the average (median) market value of COMPUSTAT universe (mean of $1,681 million; median of $88 million). The greater standard deviation of $\overline{CII_CO_f}$ ($\overline{CII_ASY_f}$) relative to that of $CII_CO_{ft} - \overline{CII_CO_f}$ ($CII_ASY_{ft} - \overline{CII_ASY_f}$) suggests that the between-firm variance of CII_CO_{ft} (CII_ASY_{ft}) dominates the within-firm one. The weighted average of the presence of long-term CIIs with a holding period more than 2 years is 4%.

The greater standard deviation of $\overline{CII_Long_f}$ relative to that of $CII_Long_{ft} - \overline{CII_Long_f}$ suggests a greater between-firm variance than the within-firm one. The weighted average of asymmetric dependence of the supplier on their buyers ($ADEP_{ft}$) is 20%. The weighted average of the presence of the same director, analysts, and auditor is 0.4%, 25%, and 21%, respectively. For other control variables, at the median level, suppliers have an institutional ownership ($INST_{ft}$) of 12%, a sales growth rate of 10%, and a debt ratio of 14%, and a net income to sales ratio of 3%. The median size in natural logarithm (age) of the suppliers is around 5.18 million dollars (11 years). All control variables except *Sales growth*$_{ft}$ have greater between-firm variances than within-firm variances.

5.2 | Effects of CIIs on supplier performance

Table 3 reports our main results. In Column 1 based on Equation (4), the coefficient on the between-firm component of CII_D_{ft} ($\overline{CII_D_f}$) is 0.09 and significantly positive ($p < .05$), indicating that suppliers with the presence of CIIs outperform peers without, consistent with Hypothesis 1. In economic terms, one standard deviation increase in $\overline{CII_D_f}$ corresponds to a rise in gross profit margin (GPM_{ft+1}) by 2 percentage points (0.26×0.09). However, the coefficient on the within-firm component of CII_D_{ft} ($CII_D_{ft} - \overline{CII_D_f}$) is not significant. Columns 2–3 present the effect of CIIs on supplier performance in groups with high and low dependence. The coefficient on $\overline{CII_D_f}$ is only significant in the high dependence group ($p < .01$), which is significantly greater than that in the low dependence group as shown in the bottom line ($p < .10$). This result supports Hypothesis 2.

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TABLE 2 Descriptive statistics

Variable	Obs.	Mean	Std.	P1	Q1	Median	Q3	P99
GPM_{ft+1}	11,052	0.24	0.91	−6.36	0.23	0.37	0.53	0.93
$\overline{GPM}_f$	11,052	0.24	0.76	−3.84	0.23	0.36	0.51	0.89
GPM_{ft+1}- $\overline{GPM}_f$	11,052	0.00	0.49	−2.03	−0.03	0.00	0.03	1.65
CII_D_{ft}	11,052	0.82	0.37	0.00	1.00	1.00	1.00	1.00
$\overline{CII_D}_f$	11,052	0.82	0.26	0.00	0.75	0.95	1.00	1.00
CII_D_{ft} - $\overline{CII_D}_f$	11,052	0.00	0.25	−0.86	0.00	0.00	0.10	0.51
CII_CO_{ft}	11,052	4.45	2.95	0.00	2.00	4.50	7.00	9.00
$\overline{CII_CO}_f$	11,052	4.45	2.31	0.00	2.67	4.50	6.36	8.50
CII_CO_{ft} - $\overline{CII_CO}_f$	11,052	0.00	1.82	−5.33	−1.00	0.00	1.04	4.20
CII_ASY_{ft}	11,052	4.45	2.95	0.00	2.00	4.50	7.00	9.00
$\overline{CII_ASY}_f$	11,052	4.45	2.11	0.00	3.00	4.71	6.00	8.67
CII_ASY_{ft} - $\overline{CII_ASY}_f$	11,052	0.00	2.05	−5.56	−1.15	0.00	1.33	4.60
CII_Long_{ft}	11,052	0.04	0.05	0.00	0.00	0.01	0.06	0.24
$\overline{CII_Long}_f$	11,052	0.04	0.04	0.00	0.00	0.03	0.06	0.18
CII_Long_{ft} - $\overline{CII_Long}_f$	11,052	0.00	0.03	−0.09	−0.01	0.00	0.01	0.12
$ADEP_{ft}$	11,052	0.20	0.18	−0.11	0.10	0.15	0.24	0.90
CDI_{ft}	11,052	0.00	0.06	0.00	0.00	0.00	0.00	0.00
$\overline{CDI}_f$	11,052	0.00	0.05	0.00	0.00	0.00	0.00	0.02
CDI_{ft} - $\overline{CDI}_f$	11,052	0.00	0.03	0.00	0.00	0.00	0.00	0.00
CAN_{ft}	11,052	0.25	0.42	0.00	0.00	0.00	0.54	1.00
$\overline{CAN}_f$	11,052	0.25	0.34	0.00	0.00	0.06	0.48	1.00
CAN_{ft} - $\overline{CAN}_f$	11,052	0.00	0.25	−0.70	−0.04	0.00	0.00	0.80
CAU_{ft}	11,052	0.21	0.39	0.00	0.00	0.00	0.10	1.00
$\overline{CAU}_f$	11,052	0.21	0.32	0.00	0.00	0.00	0.35	1.00
CAU_{ft} - $\overline{CAU}_f$	11,052	0.00	0.22	−0.61	0.00	0.00	0.00	0.75
$INST_{ft}$	11,052	0.13	0.13	0.00	0.03	0.09	0.19	0.58
$\overline{INST}_f$	11,052	0.13	0.10	0.00	0.06	0.12	0.18	0.43
$INST_{ft}$ - $\overline{INST}_f$	11,052	0.00	0.09	−0.21	−0.04	0.00	0.04	0.27
$Size_{ft}$	11,052	5.41	1.94	1.51	3.99	5.18	6.73	10.32
$\overline{Size}_f$	11,052	5.41	1.87	1.77	4.05	5.16	6.68	10.17
$Size_{ft}$ - $\overline{Size}_f$	11,052	0.00	0.54	−1.72	−0.20	0.01	0.24	1.41
$Sales\ growth_{ft}$	11,052	0.23	0.58	−0.58	−0.03	0.10	0.30	3.74
$\overline{Sales\ growth}_f$	11,052	0.23	0.33	−0.18	0.05	0.14	0.28	1.70
$Sales\ growth_{ft}$ -$\overline{Sales\ growth}_f$	11,052	0.00	0.48	−1.23	−0.17	−0.02	0.12	2.09
Age_{ft}	11,052	15.35	13.30	1.00	6.00	11.00	21.00	57.00
$\overline{Age}_f$	11,052	15.35	12.76	2.33	6.50	10.27	19.67	55.09
Age_{ft} _ $\overline{Age}_f$	11,052	0.00	3.75	−10.50	−1.83	0.00	2.00	9.78
$Debt\ ratio_{ft}$	11,052	0.19	0.20	0.00	0.01	0.14	0.31	0.93
$\overline{Debt\ ratio}_f$	11,052	0.19	0.18	0.00	0.04	0.16	0.30	0.74
$Debt\ ratio_{ft}$ - $\overline{Debt\ ratio}_f$	11,052	0.00	0.10	−0.28	−0.04	0.00	0.04	0.34
ROS_{ft}	11,052	−0.25	1.09	−7.84	−0.08	0.03	0.08	0.41
$\overline{ROS}_f$	11,052	−0.25	0.90	−4.79	−0.09	0.01	0.05	0.25
ROS_{ft} - $\overline{ROS}_f$	11,052	0.00	0.62	−2.95	−0.03	0.01	0.07	1.88

Note: See Appendix A for variable definitions.

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TABLE 3 Effects of CIIs on supplier operating performance

	(1)	High dependence (2)	Low dependence (3)	(4)
Intercept	0.35***	0.44***	0.23	0.35***
	(2.66)	(2.64)	(1.18)	(2.59)
$\overline{CII_D_f}$	0.09**	0.14***	0.04	0.19***
	(2.29)	(2.72)	(1.05)	(3.16)
CII_D_{ft} - $\overline{CII_D_f}$	0.02	0.04	−0.03	0.00
	(0.98)	(1.09)	(−1.03)	(0.14)
$\overline{CII_CO_f}$				0.02**
				(2.11)
CII_CO_{ft} - $\overline{CII_CO_f}$				0.00
				(0.79)
$\overline{CII_ASY_f}$				−0.04***
				(−3.82)
CII_ASY_{ft} - $\overline{CII_ASY_f}$				0.00
				(0.16)
$\overline{CDI_f}$	0.15	0.36	0.10	0.14
	(1.49)	(1.18)	(1.27)	(1.32)
CDI_{ft} - $\overline{CDI_f}$	−0.03	0.19	−0.02	−0.03
	(−0.76)	(0.75)	(−0.57)	(−0.68)
$\overline{CAN_f}$	−0.05	−0.05	−0.05	−0.05
	(−1.26)	(−0.97)	(−1.27)	(−1.31)
CAN_{ft} - $\overline{CAN_f}$	−0.02	−0.03	−0.01	−0.02
	(−0.83)	(−0.73)	(−0.35)	(−0.91)
$\overline{CAU_f}$	−0.02	−0.00	−0.03	−0.02
	(−0.75)	(−0.02)	(−0.99)	(−0.79)
CAU_{ft} -$\overline{CAU_f}$	0.02	0.04	0.00	0.02
	(0.71)	(0.89)	(0.04)	(0.75)
$\overline{INST_f}$	−0.08	−0.19	0.04	0.19
	(−0.84)	(−1.35)	(0.38)	(1.52)
$INST_{ft}$ - $\overline{INST_f}$	0.12	0.06	0.03	0.08
	(1.32)	(0.40)	(0.26)	(0.83)
$\overline{Size_f}$	−0.01	−0.01	−0.00	−0.01
	(−1.05)	(−1.06)	(−0.28)	(−1.39)
$Size_{ft}$ - $\overline{Size_f}$	−0.02	0.01	−0.04	−0.02
	(−0.57)	(0.33)	(−1.34)	(−0.68)
$\overline{Sales\ growth_f}$	0.21***	0.24***	0.16**	0.21***
	(3.17)	(2.97)	(1.97)	(3.12)
$Sales\ growth_{ft}$ - $\overline{Sales\ growth_f}$	0.05**	0.08**	−0.02	0.05**
	(2.20)	(2.46)	(−0.57)	(2.23)
$\overline{Age_f}$	−0.00	−0.00	−0.00	−0.00
	(−0.71)	(−0.53)	(−1.09)	(−0.62)
Age_{ft} - $\overline{Age_f}$	0.00	0.01	−0.00	0.00
	(0.49)	(1.14)	(−0.34)	(0.33)
$\overline{Debt\ ratio_f}$	0.25***	0.31***	0.12	0.25***

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TABLE 3 (Continued)

	(1)	High dependence (2)	Low dependence (3)	(4)
	(3.32)	(3.03)	(1.41)	(3.36)
$Debt\ ratio_{ft}$ - $\overline{Debt\ ratio_f}$	0.29***	0.31**	0.20	0.28***
	(3.34)	(2.35)	(1.54)	(3.33)
$\overline{ROS}_f$	0.68***	0.71***	0.53***	0.68***
	(17.03)	(17.82)	(8.35)	(17.11)
ROS_{ft} - $\overline{ROS}_f$	0.24***	0.27***	0.17***	0.24***
	(6.07)	(5.24)	(2.95)	(6.07)
Adj. R^2	0.50	0.56	0.34	0.50
Obs.	11,052	5,850	5,202	11,052
Difference in coefficient on $\overline{CII_D_f}$ between subgroups		0.10*		
		(1.86)		

Note: The regression model is based on Equation (4). *t* statistics are in parentheses; standard errors are clustered by firm. Fiscal year and industry fixed effects are controlled for. See Appendix A for variable definitions.
*$p \leq .10$.
**$p \leq .05$.
***$p \leq .01$ (two tailed).

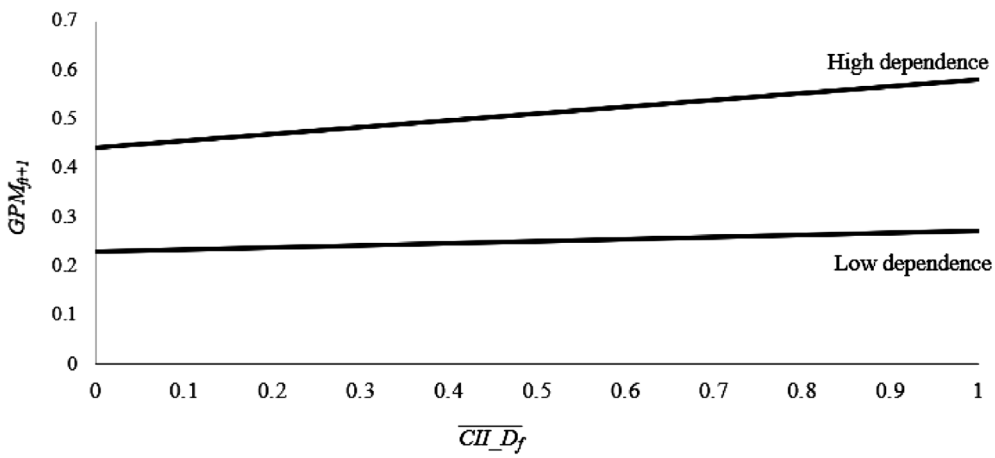

FIGURE 2 The moderating effect of $ADEP_{ft}$ on the relationship between $\overline{CII_D_f}$ and GPM_{ft+1}. This figure plots the pick-a-point simple slopes of the regression of gross profit margin (GPM_{ft+1}) on the between-firm component of CII_D_{ft} ($\overline{CII_D_f}$) at the high and low levels of asymmetric dependence of the supplier on its buyer ($ADEP_{ft}$). Firms fall in the high (low) dependence group if their asymmetric dependences on buyers are above (below) the industry median for each year. The slope for the high dependence group is significant, whereas that for low dependence group is not

Following Bauer and Curran (2005), we plot the pick-a-point simple slopes of $\overline{CII_D_f}$ at the high and low levels of $ADEP_{ft}$ in Figure 2. The steeper slope of the "high dependence" group relative to that of the "low dependence" group signifies the more pronounced effect of $\overline{CII_D_f}$ on supplier performance in the former group.

Column 4 presents the regression results after adding both the between- and within-firm components of the common ownership stakes (CII_CO_{ft}) and the asymmetric ownership stakes of CIIs (CII_ASY_{ft}) to Equation (4). The between-firm component of common ownership of CIIs ($\overline{CII_CO_f}$) is significantly and positively associated with supplier performance ($p < .05$). One standard deviation increase in $\overline{CII_CO_f}$ results in a 5 percentage point *increase* in GPM_{ft+1} (2.31×0.02). This result suggests that, after controlling for the existence of CIIs, a greater ownership stake by CIIs further enhances the operating performance of suppliers. The coefficient on $\overline{CII_D_f}$ remains significant. Further, the between-firm component of ownership asymmetry ($\overline{CII_ASY_f}$) is negatively associated with GPM_{ft+1} ($p < .01$). One standard deviation increase in

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TABLE 4 Robustness checks on the role of CIIs

	(1)	High dependence (2)	Low dependence (3)	(4)
Panel A: Two-year-ahead gross profit margin				
$\overline{CII_D_f}$	0.12**	0.15**	0.10*	0.24***
	(2.55)	(2.52)	(1.86)	(3.27)
CII_D_{ft} - $\overline{CII_D_f}$	0.03	−0.01	0.05	−0.04
	(1.02)	(−0.14)	(1.14)	(−1.04)
$\overline{CII_CO_f}$				0.02*
				(1.84)
CII_CO_{ft} - $\overline{CII_CO_f}$				0.01
				(1.17)
$\overline{CII_ASY_f}$				−0.05***
				(−3.60)
CII_ASY_{ft} - $\overline{CII_ASY_f}$				0.01*
				(1.93)
Controls	Yes	Yes	Yes	Yes
Adj. R^2	0.45	0.51	0.31	0.45
Obs.	10,400	5,465	4,935	10,400
Difference in coefficient on $\overline{CII_D_f}$ between subgroups		0.04		
		(0.66)		
Panel B: Classifying Top 20 fund families as non-CIIs				
$\overline{CII_D_f}$	0.09**	0.15***	0.05	0.20***
	(2.49)	(2.88)	(1.21)	(3.48)
CII_D_{ft} - $\overline{CII_D_f}$	0.03	0.05	−0.02	0.02
	(1.18)	(1.32)	(−0.78)	(0.43)
$\overline{CII_CO_f}$				0.01*
				(1.66)
CII_CO_{ft} - $\overline{CII_CO_f}$				0.01
				(0.83)
$\overline{CII_ASY_f}$				−0.04***
				(−3.68)
CII_ASY_{ft} - $\overline{CII_ASY_f}$				−0.00
				(−0.04)
Controls	Yes	Yes	Yes	Yes
Adj. R^2	0.50	0.56	0.34	0.50
Obs.	11,052	5,850	5,202	11,052
Difference in coefficient on $\overline{CII_D_f}$ between subgroups		0.10*		
		(1.89)		
Panel C: Alternative measure of suppliers' dependence				
$\overline{CII_D_f}$		0.15**	0.01	
		(2.12)	(0.36)	
CII_D_{ft} - $\overline{CII_D_f}$		0.00	0.03*	
		(0.09)	(1.66)	
Controls		Yes	Yes	
Adj. R^2		0.53	0.32	

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TABLE 4 (Continued)

	(1)	High dependence (2)	Low dependence (3)	(4)
Obs.		5,522	5,530	
Difference in coefficient on $\overline{CII_D_f}$ between subgroups		0.14*		
		(1.86)		
Panel D: Long-term common institutional investors				
$\overline{CII_D_f}$	0.08**	0.13**	0.03	
	(2.04)	(2.30)	(0.90)	
CII_D_{ft} - $\overline{CII_D_f}$	0.01	0.03	−0.03	
	(0.62)	(0.68)	(−1.00)	
$\overline{CII_Long_f}$	0.18	0.42	0.13	
	(0.65)	(0.94)	(0.50)	
CII_Long_{ft} - $\overline{CII_Long_f}$	0.56***	1.03***	0.02	
	(3.19)	(3.38)	(0.07)	
Controls	Yes	Yes	Yes	
Adj. R^2	0.50	0.56	0.34	
Obs.	11,052	5,850	5,202	
Difference in coefficient on CII_Long_{ft} - $\overline{CII_Long_f}$ between subgroups		1.02***		
		(2.61)		

Note: The regression model is based on Equation (4). *t* statistics are in parentheses; standard errors are clustered by firm. Fiscal year and industry fixed effects are controlled for. Results of control variables are omitted for brevity. See Appendix A for variable definitions.
*$p \leq .10$.
**$p \leq .05$.
***$p \leq .01$ (two tailed).

$\overline{CII_ASY_f}$ results in a 8 percentage point *decrease* in GPM_{ft+1} (2.11×0.04). This result suggests that CIIs with greater ownership asymmetry between the partners weaken the supplier's performance, probably because they have biased interests along the supply chain. These results support Hypotheses 3 and 4. Again, the within-firm effects of CII_CO_{ft} and CII_ASY_{ft} are insignificant.

In our research setting, we have no a priori theory to expect the between-firm effect of CIIs to differ from the within-firm effect. Rather, the differential results of between- and within-firm effects could be due to the following reasons. First, for both explanatory and dependent variables, the between-firm variance accounts for the bulk of the total variance. The limited variation of the within-firm component weakens the testing power to detect the within-firm effect. Second, omitted variables that are invariant at the firm level may introduce biases into the between-firm effect and omitted variables that change over time may bias the within-firm effect. We address the endogeneity issue in the following subsection.

Results of other communication channels are not significant. For other control variables, we find a significantly positive between- and within-firm effect for *sales growth*$_{ft}$, *Debt ratio*$_{ft}$ and ROS_{ft}. The significant coefficient on ROS_{ft} indicates the time-series correlation of firm performance.

5.3 | Robustness tests

5.3.1 | Two-year-ahead gross profit margin

Equations (3) and (4) focus on the relationship between CIIs and one-year-ahead supplier firm performance. Since it may take time for CIIs to exert influence on the partners, we use two-year-ahead gross profit margin as an alternative performance measure. The results are reported in Panel A of Table 4. The coefficient on $\overline{CII_D_f}$ is significantly positive in the full sample and both high and low dependence group. Though the coefficient is greater in the high dependence group than in the low dependence group, the difference is not significant. Consistent with Table 3, the coefficient on $\overline{CII_CO_f}$ is significantly positive and the coefficient on $\overline{CII_ASY_f}$ is significantly negative. These results confirm that the role of CIIs in improving supplier performance is not temporary.

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5.3.2 | Classifying Top 20 fund families as conventional institutional investors

Institutional ownership data in the Thomson Reuters Institutional (13F) Holdings database aggregate ownership across all funds for each institution. One may argue that different fund managers within the same institution could have opposite voting decisions on the same proxy if they attempt to maximize the benefit of their fundholders. To address this concern, we re-construct all explanatory variables after excluding Top 20 fund families because large fund families consist of more funds. Top 20 fund families are those ranked Top 20 by the average number of funds under management. We classify Top 20 fund families holding a buyer and its supplier as conventional institutional investors rather than CIIs. The results reported in Panel B of Table 4 are consistent with those in Table 3.

5.3.3 | Alternative measure of asymmetric dependence

Firms in industries with high R&D investments tend to offer products of higher complexity. Suppliers in such industries tend to invest more in relationship-specific assets, making them more dependent on their buyers. We use the median value of R&D expenditure[15] over net sales of the supplier' industry as an alternative measure of suppliers' dependence on buyers. Industry classification is based on suppliers' two-digit SIC code. Panel C of Table 4 shows that the coefficient on the between-firm component of CII_D_{ft} is only significant in the high dependence group ($p < .05$), which is significantly greater than that in the low dependence group ($p < .10$).

5.3.4 | Cross-sectional analysis for long-term institutional investors

A further review of our data indicates that, among relationship observations with CIIs, 66% have at least one CII holding at least 1% ownership in the partners. Since prior research (e.g., Brickley, Lease, & Smith, 1988) uses 1% ownership as a cutoff in determining whether active institutional investors have an incentive to monitor managers,[16] we expect that the majority of CIIs in our sample are incentivized to play a monitoring role.

Literature also indicates that long-term institutional investors have monitoring incentives (e.g., McCahery et al., 2016), whereas short-term institutional investors play an informational role (e.g., Yan & Zhang, 2009). As such, if the monitoring role of CIIs explains our results, the results should be more pronounced for long-term CIIs. Following McCahery et al. (2016), we classify CIIs as a long-term investor if their holding period of a buyer–supplier relationship is longer than 2 years. We create CII_Long_{ft}, which is a weighted indicator of a long-term CII by the ownership the CII holds in the supplier and then is weighted by the percentage of sales to the buyer to be converted to the firm level. We include both the between- and within-firm components of CII_Long_{ft} ($\overline{CII_Long_f}$ and $CII_Long_{ft} - \overline{CII_Long_f}$) in Equation (4).

Panel D of Table 4 shows that in Column 1, the coefficient on the within-firm component of CII_Long_{ft} (CII_Long_{ft} - $\overline{CII_Long_f}$) is positive and significant, while the coefficient on CII_D_{ft} - $\overline{CII_D_f}$ is insignificant. Columns 2–3 demonstrate that the effect of long-term CIIs is more pronounced for supplier firms with a greater dependence on buyers.

5.3.5 | Effects of CIIs on buyers and supply chain performance

Our study focuses on the effect of CIIs on supplier performance. However, to gain a more complete understanding of the role of CIIs in the entire supply chain, we further investigate the effects of CIIs on buyer performance and the combined supply chain performance. Our additional analyses reveal that the presence of CIIs improves the industry-adjusted gross profit margin for the entire supply chains. However, the effect on buyer's gross profit margin alone falls short of statistical significance.[17,18] Although the logic of our hypotheses also applies to the buyer, the latter is not the focus of our study.

5.4 | Addressing endogeneity

The endogenous nature of institutional ownership has been well documented in the literature. In our research setting, institutional investors may choose to invest in both the supplier and buyer based on certain characteristics of the partners or the relationships. In this section, we address the endogeneity concern in two ways. First, we utilize the propensity score matching (PSM) method. Second, we restrict the sample to CIIs that existed only after the formation of a buyer–supplier relationship.

5.4.1 | PSM analysis

To apply the PSM method, we estimate the probability of the presence of CIIs (propensity score) in the first stage.

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TABLE 5 Endogeneity tests

	Propensity score matching		CIIs existed after the formation of buyer–supplier relationship	
	(1)	(2)	(3)	(4)
$\overline{CII_D_f}$	0.09**	0.18***	0.07*	0.24**
	(2.05)	(2.90)	(1.69)	(2.24)
CII_D_{ft} - $\overline{CII_D_f}$	0.02	0.00	0.02	0.01
	(0.95)	(0.03)	(0.85)	(0.13)
$\overline{CII_CO_f}$		0.02**		0.03
		(2.06)		(1.18)
CII_CO_{ft} - $\overline{CII_CO_f}$		0.01		−0.01
		(0.86)		(−0.93)
$\overline{CII_ASY_f}$		−0.04***		−0.06**
		(−3.81)		(−2.28)
CII_ASY_{ft} - $\overline{CII_ASY_f}$		0.00		0.01
		(0.32)		(1.29)
Controls	Yes	Yes	Yes	Yes
Adj. R^2	0.51	0.51	0.20	0.21
Obs.	10,256	10,256	5,273	5,273

Note: The regression model is based on Equation (4). *t* statistics are in parentheses; standard errors are clustered by firm. Fiscal year and industry fixed effects are controlled for. Results of control variables are omitted for brevity. See Appendix A for variable definitions.
*$p \leq .10$.
**$p \leq .05$.
***$p \leq .01$ (two tailed).

The determinants include the communication channels between the partners, characteristics of the partnership, fundamentals of the partners, and industry and time effects.[19] For each partnership-year in the treatment sample (relationship-years with CIIs), we choose one partnership-year in the control sample (relationship-years without CIIs) that has the closest propensity score and the absolute difference in propensity score is less than 0.1. Partnership-years with an available match and their matching counterparties consist of our new sample at the relationship-year level. We then convert the relationship-level variables into firm-level ones by using the weighted average measures. The weight is the percentage of sales to the buyer.

In the second stage, we run Equation (4) with this new sample and report the results in Columns 1–2 of Table 5. Column 1 shows that the coefficient on $\overline{CII_D_f}$ is significantly positive.[20] Column 2 shows that the between-firm component of common ownership ($\overline{CII_CO_f}$) is positively associated with supplier performance, while the between-firm component of ownership asymmetry ($\overline{CII_ASY_f}$) is negatively associated with supplier performance. The PSM analysis generates results consistent with our main findings.

5.4.2 | CIIs existed after the formation of buyer–supplier relationships

To mitigate the concern that CIIs may guide the buyer to choose a supplier with better performance, we restrict our relationship-level sample to CIIs that exist after the formation of a buyer–supplier relationship and redo our main analysis. In such a sample, CIIs have no chance to influence a buyer's selection of supplier. Again, we convert relationship-level variables into firm-level ones by using the weighted average value of these variables. The weight is the percentage of sales to the buyer. The results shown in Columns 3–4 of Table 5 remain qualitatively similar to those in Table 3. The collective evidence from Table 5 strengthens the causal inferences.

5.5 | Tests of market performance of suppliers

To the extent that CIIs aim to maximize their portfolio value, it would be worthwhile to examine the effect of CIIs on supplier market performance. Moreover, while financial performance captures firm performance

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TABLE 6 Effects of CIIs on supplier market performance

	(1)	High dependence (2)	Low dependence (3)	(4)
Intercept	−0.02	−0.02	−0.01	−0.01
	(−1.31)	(−1.23)	(−0.52)	(−0.54)
ΔCII_D_{ft}	0.04	0.10**	−0.06	0.04
	(1.15)	(2.25)	(−1.21)	(1.32)
ΔCII_CO_{ft}				0.01**
				(2.34)
ΔCII_ASY_{ft}				−0.01***
				(−2.87)
ΔCDI_{ft}	−0.08	0.33***	−0.14	−0.08
	(−0.81)	(2.67)	(−1.28)	(−0.79)
ΔCAN_{ft}	0.03	0.01	0.05	0.03
	(1.18)	(0.26)	(1.51)	(1.18)
ΔCAU_{ft}	−0.00	−0.01	0.01	−0.00
	(−0.03)	(−0.20)	(0.11)	(−0.01)
$\Delta INST_{ft}$	1.12***	1.11***	1.14***	1.16***
	(9.24)	(6.59)	(7.51)	(8.97)
$\Delta E_{ft}/M_{f,t-1}$	0.65***	0.61***	0.72***	0.65***
	(11.96)	(8.89)	(8.51)	(11.98)
$\Delta A_{ft}/M_{f,t-1}$	0.40***	0.45***	0.34***	0.40***
	(10.89)	(9.49)	(6.39)	(10.88)
$\Delta RD_{ft}/M_{f,\ t-1}$	−0.25	−0.38	0.04	−0.26
	(−0.57)	(−0.62)	(0.06)	(−0.60)
$\Delta I_{ft}/M_{t-1}$	−4.57***	−5.08***	−3.90***	−4.56***
	(−7.58)	(−6.21)	(−4.67)	(−7.58)
$\Delta D_{ft}/M_{t-1}$	1.26*	1.60	0.84	1.30*
	(1.67)	(1.37)	(0.84)	(1.71)
$C_{f,t-1}/M_{t-1}$	0.34***	0.37***	0.30***	0.34***
	(5.92)	(4.84)	(3.71)	(5.94)
L	−0.31***	−0.34***	−0.28***	−0.31***
	(−8.99)	(−7.48)	(−5.90)	(−9.00)
NF/M_{t-1}	0.12*	0.03	0.26**	0.12*
	(1.69)	(0.29)	(2.46)	(1.67)
Adj. R^2	0.20	0.19	0.21	0.20
Obs.	7,037	3,796	3,238	7,037
Difference in coefficient on ΔCII_D_{ft} between subgroups		0.15**		
		(2.45)		

Note: t statistics are in parentheses; standard errors are clustered by firm. Fiscal year and industry fixed effects are controlled for (omitted for brevity). See Appendix A for variable definitions.
*$p \leq .10$.
**$p \leq .05$.
***$p \leq .01$ (two tailed).

pertaining to a specific period, market performance reflects the perception of the market about the firm's future performance. Therefore, using market performance as an alternative performance measure mitigates the concern that the delayed effect of CIIs could make the within-firm components of CIIs insignificant. We follow Faulkender and

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Wang (2006)'s excess return model to detect the *change-in-changes* effect. The dependent variable is excess return during the fiscal year, calculated as the realized stock return minus the value-weighted return of the Fama–French size-BE/ME portfolio to which the firm belongs ($r_{ft} - R^B$). ΔCII_D_{ft} is the change in the presence of CIIs from $t-1$ to t. The variable is weighted by the percentage of sales to the buyer to be converted into the firm level. ΔCII_CO_{ft} is the decile rank of weighted average change in common ownership from $t-1$ to t, as defined in Section 3; ΔCII_ASY_{ft} is the decile rank of weighted average change in ownership asymmetry of CIIs from $t-1$ to t. Since the between-firm component of a variable does not change over time, the change of the variable is equivalent to the change of its within-firm component.

We report the results in Table 6. In Column 1, the coefficient on ΔCII_D_{ft} is insignificant. However, Column 2 shows that there is a positive and significant association ($p < .05$) between ΔCII_D_{ft} and excess returns for suppliers that are more dependent on their buyers, while the coefficient on ΔCII_D_{ft} is insignificant in the low dependence group (Column 3). The difference in coefficients on ΔCII_D_{ft} between the two groups is significant ($p < .05$) as shown in the bottom row. Column 4 indicates that the change in common ownership (ΔCII_CO_{ft}) is significantly and positively associated with supplier excess returns. The change in ownership asymmetry (ΔCII_ASY_{ft}) is significantly and negatively associated with supplier excess returns. Overall, the results from the market return analysis corroborate the value-enhancing role of common investors.

5.6 | Tests of trade credits offered by suppliers

If CIIs enhance collaboration and coordination between partners and/or mitigate opportunism, we should observe a negative association between CIIs and trade receivables of suppliers. Table 7 presents the analysis. The dependent variable is the industry-adjusted trade receivables deflated by total sales of suppliers. Industry-level trade receivables deflated by total sales is the median value of trade receivables over total sales among all COMPUSTAT firms having the same two-digit SIC code. Control variables are the same as those in Equation (4). The coefficient on the between-firm component of CII_D_{ft} ($\overline{CII_D_f}$) is significantly negative in Column 1, indicating that suppliers with the presence of CIIs have lower trade receivables than those without. In Column 2, the coefficient on the between-firm component of CII_CO_{ft} ($\overline{CII_CO_f}$) is significantly negative, whereas the coefficient on $\overline{CII_ASY_f}$ is significantly positive. These results suggest that suppliers with a higher level of common ownership or lower level of ownership asymmetry have lower trade receivables, signaling an enhanced relationship commitment. To the extent that a reduction of trade credits corresponds to a quicker receivable turnover, it is a possible mechanism through which CIIs enhance supplier performance.

6 | COMPARATIVE ANALYSIS OF CIIS AND DIRECT EQUITY LINKS

The literature documents the effects of direct equity ownership (i.e., a focal firm–supplier cross-ownership) on relationship performance (e.g., Allen & Phillips, 2002; Dyer, 1997; Fee et al., 2006). In this section, we compare the effects of the two types of ownership, indirect cross-ownership (i.e., common institutional ownership) and direct equity links, on supplier performance.

We collect information on equity stakes held by buyers in suppliers (or vice versa) mainly from the SEC EDGAR system. Since EDGAR's coverage starts in 1994, we search

TABLE 7 Tests on a mechanism of CIIs' effects on supplier performance

	Trade receivables	
	(1)	(2)
$\overline{CII_D_f}$	−0.01**	−0.01
	(−2.00)	(−0.60)
CII_D_{ft} - $\overline{CII_D_f}$	−0.00	−0.00
	(−0.80)	(−0.48)
$\overline{CII_CO_f}$		−0.00***
		(−3.51)
CII_CO_{ft} - $\overline{CII_CO_f}$		−0.00
		(−0.84)
$\overline{CII_ASY_f}$		0.00**
		(2.00)
CII_ASY_{ft} - $\overline{CII_ASY_f}$		0.00
		(0.64)
Controls	Yes	Yes
Adj. R^2	0.01	0.02
Obs.	10,780	10,780

Note: Control variables are those in Equation (4). Control variable results are omitted for brevity. t statistics are in parentheses; standard errors are clustered by firm. See Appendix A for variable definitions.
*$p \leq .10$.
**$p \leq .05$.
***$p \leq .01$ (two tailed).

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TABLE 8 Comparative analysis of CIIs and direct equity links

Panel A: Descriptive statistics for direct equity links							
	N	Mean	P1	Q1	Median	Q3	P99
$CII_D_{i,t}$	14,708	82.52%	0	1	1	1	1
$DL_D_{i,t}$	14,708	1.84%	0	0	0	0	1
Direct ownership$_{i,t}$	270	11.19%	1.10%	6.10%	8.11%	17.90%	24.10%

Panel B: Effects of direct links and CIIs on supplier profit margin			
	(1)	(2)	(3)
$\overline{DL_D_f}$	−1.19***	−1.19***	−0.33**
	(−3.19)	(−3.20)	(−2.13)
DL_D_{ft} - $\overline{DL_D_f}$	0.09	0.08	0.19
	(0.32)	(0.30)	(0.75)
$\overline{CII_D_f}$		−0.00	0.08**
		(−0.06)	(2.20)
CII_D_{ft} - $\overline{CII_D_f}$		0.03	0.02
		(1.02)	(0.90)
Controls	No	No	Yes
Adj. R^2	0.02	0.02	0.50
Obs.	11,052	11,052	11,052

Note: t statistics are in parentheses; standard errors are clustered by firm. Only the results of variables of interest are reported for brevity. Industry and Year fixed effects are controlled for in Panel B. The regression model of Panel B is based on Equation (4). See Appendix A for variable definitions.
*$p \leq .10$.
**$p \leq .05$.
***$p \leq .01$ (two tailed).

for the information on direct equity links online for relationship-years before 1994 (873 firm-pairs).[21]

Panel A of Table 8 provides the descriptive statistics for direct equity links at the relationship level. $DL_D_{i,t}$ is 1 for relationship i in year t with direct equity links (one party holds stock of the other), and 0 otherwise. As noted earlier, direct equity links are rare, existing in only 1.84% of our relationship-years sample. The frequency is substantially smaller than that of the presence of CIIs ($CII_D_{i,t}$) which is 82.52% in our sample. For relationship-years with direct equity links, the average percentage of shares the buyer holds in the supplier is around 11.19%.[22]

Panel B of Table 8 presents the results of associations of direct equity links and CIIs with supplier gross profit margin at the firm level. We weight the presence of direct equity links by the percentage of sales to the buyer to convert the variable into the firm level (DL_D_{ft}). We then decompose DL_D_{ft} into between- and within-firm components: $\overline{DL_D_f}$ and $DL_D_{ft} - \overline{DL_D_f}$. In both Columns 1 and 2 without control variables, the coefficients of $\overline{DL_D_f}$ are significantly negative. The coefficient on $\overline{CII_D_f}$ in Column 2 is insignificant. However, the coefficient on $\overline{CII_D_f}$ becomes positive and significant after adding all control variables in Column 3, while the coefficient on $\overline{DL_D_f}$ remains significantly negative. These results indicate that, while direct equity ownership may contribute to total supply chain value, it adversely affects supplier gross profit margin. In contrast, CIIs create value for dependent suppliers.

7 | DISCUSSION

Our research offers a novel extension to prior studies of embeddedness that have focused on extended buyer–supplier trade networks (Dong et al., 2015; Kim, 2014; Kim, 2017) and nation-based networks (Cao & Lumineau, 2015; Dong et al., 2016; Handley & Angst, 2015; Sheng et al., 2018; Wang et al., 2016). Using social network theory and power dependence theory to derive our expectations, we employ an econometric approach to uncover supporting evidence of unobservable mechanisms consistent with our predictions (Makadok, Burton, & Barney, 2018; Sutton & Staw, 1995).[23] Our findings suggest that CIIs enhance both the operating and market performance of dependent suppliers. These results are consistent with the

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proposition that CIIs with higher common ownership have greater incentives to strengthen joint optimization and coordination among supply chain partners, while curbing opportunism. On the other hand, higher CII ownership asymmetry between partners degrades supplier performance. This evidence further supports the notion of CIIs' influence, as it suggests that CIIs' ownership asymmetry reduces their incentives to encourage buyer–supplier coordinative actions, and may lessen CII constraints on buyer opportunism. Also supporting the role of CIIs is the post hoc finding that the positive effect of CIIs on supplier performance is more pronounced for long-term CIIs. Presumably, longer tenure provides greater opportunity for influential actions such as those mentioned above, and more time for buyer–supplier pairs to develop common operational norms. An interesting question for future research is whether the apparent benefits of long-term CIIs degrade over time due to hold-up, lock-in, or other rigidities associated with focused commitments in buyer–supplier relationships (Dong et al., 2015; Kim & Zhu, 2018; Swink & Zsidisin, 2006).

Our analysis also identifies one potential operational indicator of enhanced relationship commitment: decreased trade credits. Suppliers' receivables are reduced in the presence of CIIs, pointing to influential roles of CIIs in reducing perceived risk and coercion in the buyer–supplier relationship (Lee & Stowe, 1993; Smith, 1987; Van Horen, 2007). The study of antecedents and effects of trade credit is emerging as a popular topic for supply chain management researchers. Studies including Lee, Zhou, and Wang (2018) and Kovach, Swink, and Rodrigues (2019) indicate that buyers' profits decrease with rising trade credits; however, these effects are moderated by competition, dependence, financial stability, and other factors. Future research might examine these effects from an information processing perspective considering how CIIs and other social network structures may serve as moderators of frictions that produce non-optimal trade-credit agreements.

Lastly, this is the first study to compare the influences of corporate direct equity ownership and indirect ownership through CIIs on supplier performance in vertical relationships. Compared with direct equity links, CIIs are more than 40 times as common in supply chains (in our sample). After controlling for direct equity links, the positive impacts of CIIs continue to manifest in the models we estimated. In contrast, the association of direct equity links with supplier profitability is strongly negative. The literature suggests that direct equity ownership creates value by promoting greater supply chain collaboration and longer relationships (e.g., Allen & Phillips, 2002; Dyer, 1997; Fee et al., 2006). Allen and Phillips (2002) find that the effect of equity purchase announcements is significantly positive on supplier profitability in high R&D industries, but it is significantly negative in low R&D industries. The presence or absence of alliances and joint ventures also moderates the effects in their findings. Together, our results and the findings of prior research suggest that the effects of direct equity ownership on buyer–supplier relationships are complex and likely moderated by other relational factors. For example, Dyer (1996) discusses benefits of direct equity stakes, but primarily in Japanese firms, where norms of obligation associated with equity affiliation are high, and significant obstacles to equity divestment exist (Ahmadjian & Oxley, 2005). In such a circumstance, corporate equity ownership can act as a type of hostage arrangement that induces commitment by both parties (Ahmadjian & Oxley, 2005; Williamson, 1985). However, Ahmadjian and Oxley (2005) argue that if the size of investment is small and financial markets are liquid (as in the United States), the impact of equity ownership may be negligible. Our results seem to suggest that in some circumstances, direct equity ownership might even serve as a means by which a buyer expropriates profits from its supplier (Lanier Jr et al., 2010). The complexity of these issues points to the need for continued study that develops a more complete theory of direct equity ownership as a governance mechanism.

We acknowledge that this study is subject to several limitations. First, we use secondary data to address relationships that are difficult to observe in practice directly. We offer theory-based explanations for hypothesized effects. However, qualitative research efforts might produce a richer understanding of the roles and mechanisms of CIIs who contribute to the governance of buyer–supplier relationships, while additional quantitative studies using primary data might provide more direct tests of theorized relationships. Future research may use surveys or other methods to directly capture the role of CIIs in buyer–supplier relationship and supplier performance. Second, our study is limited in its examination of the mechanisms that CIIs may use to strengthen collaboration and coordination and constrain opportunism among buyers and their suppliers. We conducted post hoc analyses that also revealed that (a) buyer–supplier relationships with CIIs are more likely to have connected directors from the same employer and (b) buyers with CIIs are more likely to receive shareholder proposals, which encourage suppliers to increase corporate social responsibility activities. For brevity, these results are not reported. Empirically, it is difficult to directly and explicitly capture underlying mechanisms of CIIs' influence. However, future research might explore additional mechanisms, again using different research methods. Third, the presence of CIIs tends to be endogenous.

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Though we use various empirical methods to address endogeneity issue, we still cannot fully preclude the effect of endogeneity of CIIs on our results. Future research might search for some exogenous shock to the presence of CIIs to mitigate the endogeneity issue. Finally, we identify decreased trade receivables as a mechanism through which CIIs increase supplier performance. There might be some omitted correlated mechanisms through which CIIs affect supplier–buyer relationships. Future research can seek to identify the alternative mechanisms and improve our understanding of the role of CIIs in supply chains.

8 | CONCLUSION

Buyer–supplier relationships are embedded in complex networks; and they are potentially influenced by other trade partners, owners (direct and institutional), regulators, activist groups, analysts, industry associations, and perhaps even other entities. Our study addresses a heretofore scarcely studied source of social influence, CIIs. The positive associations of both the presence and intensity of CIIs with supplier performance appear to be quite consistent, robust, and distinct from the influences of direct equity links in buyer–supplier dyads. Overall, these findings extend the application of social network theory in supply chain management beyond current considerations of exchange partnerships and national institutions. They demonstrate that institutional investors also play important roles in relational governance, while highlighting a number of interesting questions and opportunities for future research.

ACKNOWLEDGMENT

We gratefully acknowledge insightful suggestions from the editors and review team, which substantively improved the earlier version of this manuscript. We appreciate helpful comments from Wenming Wang, Bill Wempe, Donghui Wu, Weiqiang Tan, Steven Xu Zhang, and the seminar participants at Zhongnan University of Economics and Law, Hong Kong Baptist University, University of Macau, and Hang Seng University of Hong Kong. All coauthors have made equal contribution to the paper. Zhang acknowledges financial support from the National Natural Science Foundation of China (No. 71702188). Haw acknowledges that parts of his work were conducted during his visit to Hang Seng University of Hong Kong. The authors alone are responsible for all limitations and errors that may relate to the paper.

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ENDNOTES

[1]Following prior studies (e.g., Klein & Zur, 2009), we define institutional investors as investment companies (e.g., mutual fund management companies) and their managers as well as independent investment advisors (e.g., asset management companies, investment banks, brokers, private wealth management companies) who invest in securities for its own accounts or on behalf of others (Agarwal, Jiang, Tang, & Yang, 2013). Therefore, neither an individual shareholder nor a corporation is considered as an institutional investor. A supplier–buyer relationship is deemed to have a common institutional investor in year *t* if an institutional investor simultaneously holds shares of the buyer and the supplier for more than two quarters of year *t*.

[2]The portfolio value for CIIs can be regarded as their ownership in supplier multiplied by supplier profit plus their ownership in buyer multiplied by buyer profit.

[3]This is the number of relationship observations before excluding those where the buyer and supplier are in a business group or where the buyer or supplier is a financial or utility firm.

[4]The majority of prior research on U.S. supplier–buyer relationships uses the same data source (for example, Cohen & Frazzini, 2008; Fee et al., 2006).

[5]The SEC EDGAR system provides company filings beginning in 1994. For relationship-years before 1994, we manually search for the firm-pairs on the website to find the shareholding information of one partner in the other.

[6]Please refer to Section 4 for the definitions of between- and within-firm components of each variable.

[7]Since the institutional identifier (MGRNO) in the database is not unique or permanent over time, we check whether the same MGRNO represents different institutions and whether different MGRNOs indicates the same institution by searching the SEC EDGAR system. In the former situation, we assign a different MGRNO to the latter institution. In the latter situation, we assign the same institution a uniform MGRNO, the smaller one of the two MGRNOs. When we cannot determine whether two institutions are different, we treat them as different ones.

[8]We use supplier ownership as the weight because our study focuses on the effect of CIIs on supplier performance and a CII's ownership in a supplier captures her incentive and ability to increase supplier performance. A CII holding a low ownership in a supplier has little incentive to improve supplier performance even if she holds a high ownership in the buyer.

[9]Extant research such as Patatoukas (2012) and Radhakrishnan et al. (2014) uses the same method to convert relationship-level variables to the supplier firm level.

[10]Patatoukas (2012) examines the relationship between customer concentration and firm profitability. He finds that firm size and sales growth are positively associated with supplier gross profit margin, whereas firm age and leverage are negatively associated with supplier gross profit margin. Our results on these control variables are generally consistent with his findings.

[11]Hall et al. (2010) state that one-year-ahead dependent variable mitigates the simultaneity bias. Callen and Fang (2013) argue that using one-year-ahead stock price crash risk alleviates the reverse causality concern between institutional investor stability and stock price crash risk. Balafas and Florackis (2014) examine the

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relationship between CEO compensation and firm value and argue that using lagged independent variables alleviates the reverse causality issue. Qian and Steiner (2017) examine the relationship between international reserves and the external debt maturity and state that using lagged independent variables helps to address the issue of reverse causality and potential endogeneity.

[12]Shaver (2019) demonstrates that the subsample analysis is better than analysis using interaction terms in capturing the differential effects of the explanatory variables in a fixed-effect model with a third variable.

[13]Intraclass correlation coefficient (ICC) is the ratio of between-firm variance to the total variance of the variable (between-firm variance plus within-firm variance).

[14]Table 2 reports descriptive statistics of the decile ranks of CII_CO_{ft} and CII_ASY_{ft}.

[15]Missing R&D expenditure is set to 0.

[16]SEC officially states that "*To be eligible to submit a proposal under Rule 14a-8, a shareholder must, among other things, provide documentation evidencing that the shareholder has continuously held at least $2,000 in market value, or 1%, of the company's securities entitled to be voted on the proposal at the shareholder meeting for at least 1 year as of the date the shareholder submits the proposal*" (https://www.sec.gov/interps/legal/cfslb14g.htm).

[17]We measure industry-adjusted supply chain performance as follows:

$$\frac{(\textit{buyer net sales} \times \textit{percentage of cost of goods sold from the supplier} - \textit{supplier cost of goods sold} \times \textit{percentage of sales to the buyer})}{\textit{buyer net sales} \times \textit{percentage of cost of goods sold from the supplier}} - \frac{(\textit{industry gross profit margin of supplier} \times \textit{industry sales of supplier} + \textit{industry gross profit margin of buyer} \times \textit{industry sales of buyer})}{\textit{industry sales of supplier} + \textit{industry sales of buyer}}$$

where industry gross profit margin and sales are those of a firm in the industry with the median-level gross profit margin.

[18]The insignificant result of the buyer performance is likely due to an inherent feature of our dataset. As firms are required to disclose their major buyers but not major suppliers (SFAS 14 and SFAS 131), our sample has comprehensive information about major buyers, but limited information about major suppliers. As a result, the buy-supplier relationships captured in our dataset are more accurate to the suppliers than to the buyers, which can more effectively detect any effect of CIIs on the suppliers than the buyers. The results on buyers and the combined supply chains are available upon request.

[19]Proxies for the communication channels between the partners include the presence of common analyst, common director, and common auditor, whether the partners are in the same industry, and whether the partners are located in the same city. Proxies for the characteristics of a partnership include the percentage of supplier sales to the buyer and the percentage of buyer cost of goods sold from the supplier. We include various proxies for the fundamentals of the partners as follows: the average firm size (natural logarithm of total asset) of the partners, and the average lagged leverage (long-term debt divided by total asset), lagged ROA (earnings before extraordinary item divided by total asset) and tangibility ratio (capital expenditures divided by total asset) of the partners. As proxies for the industry effects, we use the average industry percentage of firms with CIIs of the partners and the average industry competition (Herfindahl–Hirschman index based on sales) of the partners.

[20]We conduct the subsample analyses and find that the coefficient on $\overline{CII_D_f}$ is only significant in the high dependence group ($p < .05$).

[21]We use names of the supplier and buyer as keywords to search on Google to determine whether there are news or articles mentioning the shareholding information between the two firms. If we cannot find any information, we treat the firm-pairs as one without direct equity links.

[22]Because only shareholders that hold more than 5% of shares are required to file the change of their shareholdings, we can only identify the ownership of blockholders via EDGAR System. It is rare that supplier block-holds the stocks of the buyer.

[23]Some scholars argue that theoretical mechanisms are not captured in econometric models (see Astbury & Leeuw, 2010; Mahoney, 2001).

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How to cite this article: Cheung Y-L, Haw I-M, Hu B, Swink M, Zhang W. Common institutional investors and supplier performance in supply chains. *J Oper Manag*. 2020;66:670–696. https://doi.org/10.1002/joom.1089

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APPENDIX A. VARIABLE DEFINITIONS

Variables	Definition
$ADEP_{ft}$	Sales dependence minus cost dependence, where sales dependence is calculated as sales to the buyer divided by total sales of the supplier, and cost dependence is calculated as purchases from the supplier divided by the total cost of goods sold of the buyer. The variable is weighted by the percentage of sales to the buyer to be converted into the firm level.
$AdjARS_{ft+1}$	Trade receivables over net sales minus the industry median of trade receivables over net sales at $t + 1$.
Age_{ft}	Firm age.
$\overline{Age_f}$	The between-firm component of Age_{ft}, which is the average Age_{ft} across all fiscal years for each firm.
$C_{f,t-1}$	Cash plus marketable securities in year $t - 1$, adjusted for inflation to 2011.
CAN_{ft}	1 if the relationship has a common analyst, and 0 otherwise. The variable is weighted by the percentage of sales to the buyer to be converted into the firm level.
$\overline{CAN_f}$	The between-firm component of CAN_{ft}, which is the average CAN_{ft} across all fiscal years for each firm.
CAU_{ft}	1 if the relationship has a common auditor at the audit firm level, and 0 otherwise. The variable is weighted by the percentage of sales to the buyer to be converted into the firm level.
$\overline{CAU_f}$	The between-firm component of CAU_{ft}, which is the average CAU_{ft} across all fiscal years for each firm.
CDI_{ft}	1 if the relationship has a common director, and 0 otherwise. The variable is weighted by the percentage of sales to the buyer to be converted into the firm level.
$\overline{CDI_f}$	The between-firm component of CDI_{ft}, which is the average CDI_{ft} across all fiscal years for each firm.
CII_ASY_{ft}	The weighted average asymmetric institutional ownership that is calculated as the absolute difference between supplier ownership and buyer ownership for each CII at the end of the last quarter of her shareholding in year t. The weight is the ownership the CII holds in the supplier. It is 0 for buyer–supplier relationships without CIIs. The variable is weighted by the percentage of sales to the buyer to be converted into the firm level. We use its decile ranks for empirical tests.
$\overline{CII_ASY_f}$	The between-firm component of CII_ASY_{ft}, which is the average CII_ASY_{ft} across all fiscal years for each firm.
CII_CO_{ft}	The weighted average common institutional ownership that is calculated as the lower between supplier ownership and buyer ownership for each CII at the end of the last quarter of her shareholding in year t. The weight is the ownership the CII holds in the supplier. It is 0 for buyer–supplier relationships without CIIs. The variable is weighted by the percentage of sales to the buyer to be converted into the firm level. We use its decile ranks for empirical tests.
$\overline{CII_CO_f}$	The between-firm component of CII_CO_{ft}, which is the average CII_CO_{ft} across all fiscal years for each firm.
CII_D_{ft}	1 if the relationship has a common institutional investor who simultaneously holds the stock of the supplier and buyer for more than two quarters in the fiscal year, and 0 otherwise. The variable is weighted by the percentage of sales to the buyer to be converted into the firm level.
$\overline{CII_D_f}$	The between-firm component of CII_D_{ft}, which is the average CII_D_{ft} across all fiscal years for each firm.
CII_Long_{ft}	The weighted indicator of a long-term CII within a buyer–supplier relationship. The weight is the ownership the CII holds in the supplier. CIIs are long-term investors if their holding period of a buyer–supplier relationship is longer than 2 years. The variable is weighted by the percentage of sales to the buyer to be converted into firm level.
$\overline{CII_Long_f}$	The between-firm component of CII_Long_{ft}, which is the average CII_Long_{ft} across all fiscal years for each firm.
D_{ft}	Dividends, adjusted for inflation to 2011.
$Debt\ ratio_{ft}$	The ratio of total debt over total assets.
$\overline{Debt\ ratio_f}$	The between-firm component of $Debt\ ratio_{ft}$, which is the average $Debt\ ratio_{ft}$ across all fiscal years for each firm.
$Direct\ ownership_{it}$	The stock ownership of one partner holds in the other.
DL_D_{ft}	1 if one partner holds the stock of the other, and 0 otherwise. The variable is weighted by the percentage of sales to the buyer to be converted into firm level.

(Continues)

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Variables	Definition
$\overline{DL_D_f}$	The between-firm component of DL_D_{ft}, which is the average DL_D_{ft} across all fiscal years for each firm.
E_{ft}	Earnings before extraordinary items, adjusted for inflation to 2011.
FYEAR	Indicators for every fiscal year except the base year.
GPM_{ft+1}	Gross profit margin at $t+1$, calculated as (net sale − cost of goods sold)/net sale.
High dependence$_{ft}$	1 if $ADEP_{ft}$ is above the median of all firms in the same industry and year, and 0 otherwise.
I_{ft}	Interest expense, adjusted for inflation to 2011.
IND	Indicators for every industry based on two-digit SIC code except the base industry.
Industry R&D	Industry-level R&D expenditures divided by industry-level total assets. Industries are classified by two-digit SIC codes. Missing value of R&D expenditures is treated as 0.
$INST_{ft}$	Total ownership of all institutional investors (including both CIIs and non-CIIs) at the quarter ending at or before the fiscal year end date.
$\overline{INST_f}$	The between-firm component of $INST_{ft}$, which is the average $INST_{ft}$ across all fiscal years for each firm.
L_{ft}	Total debt/(total debt + market value of equity).
M_{ft}	Market value of equity (common shares outstanding * price at the end of the fiscal year).
NF_{ft}	Total equity issuance minus repurchases plus debt issuance minus debt redemption.
R^B	Value-weighted return of the Fama–French size-BE/ME portfolio to which the firm belongs.
r_{ft}	Realized stock return during fiscal year t.
ROS_{ft}	Net income divided by net sales.
$\overline{ROS}_f$	The between-firm component of ROS_{ft}, which is the average ROS_{ft} across all fiscal years for each firm.
Sales growth$_{ft}$	The growth rate of net sales, calculated as (net sale at t − net sale at $t-1$)/net sale at t.
$\overline{Sales\ growth_f}$	The between-firm component of *Sales growth*$_{ft}$, which is the average *Sales growth*$_{ft}$ across all fiscal years for each firm.
$Size_{ft}$	Natural logarithm of total assets.
$\overline{Size_f}$	The between-firm component of $Size_{ft}$, which is the average $Size_{ft}$ across all fiscal years for each firm.
ΔA_{ft}	Change in inflation adjusted total assets from $t-1$ to t.
ΔCAN_{ft}	Change in the status of common analysts between the supplier and buyer from $t-1$ to t. The variable is weighted by the percentage of sales to the buyer to be converted into the firm level.
ΔCAU_{ft}	Change in the status of common auditors between the supplier and buyer from $t-1$ to t. The variable is weighted by the percentage of sales to the buyer to be converted into the firm level.
ΔCDI_{ft}	Change in the status of common directors between the supplier and buyer from $t-1$ to t. The variable is weighted by the percentage of sales to the buyer to be converted into the firm level.
ΔCII_ASY_{ft}	Change in ownership asymmetry of CIIs from $t-1$ to t. The calculation of ownership asymmetry is shown in Section 5.5. The variable is weighted by the percentage of sales to the buyer to be converted into the firm level. We then use its decile ranks for empirical tests.
ΔCII_CO_{ft}	Change in common ownership of CIIs from $t-1$ to t. The calculation of common ownership is shown in Section 5.5. The variable is weighted by the percentage of sales to the buyer to be converted into the firm level. We then use its decile ranks for empirical tests.
ΔCII_D_{ft}	Change in the presence of CIIs from $t-1$ to t. The variable is weighted by the percentage of sales to the buyer to be converted into the firm level.
ΔD_{ft}	Change in D_{ft} from $t-1$ to t.
ΔE_{ft}	Change in E_{ft} from $t-1$ to t.
$\Delta INST_{ft}$	Change in $INST_{ft}$ from $t-1$ to t.
ΔI_{ft}	Change in I_{ft} from $t-1$ to t.
ΔRD_{ft}	Change in inflation adjusted R&D expenditures from $t-1$ to t.

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